Credit Card Fraud Detection

Logistic Regression

ENTRY NUMBER	TEAM MEMBER NAME	ROLE
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GitHub Repository.
https://github.com/Krxna04/AI101

Predicting Whether a Credit Card Transaction Is Fraudulent

Problem Statement

Credit card fraud causes financial losses to customers and financial institutions. But fraudulent transactions are difficult to distinguish from legitimate transactions. Machine learning can help identify potentially fraudulent transactions.

Data Source

Credit Card Fraud Detection (KaggleHub)

Dataset Overview

Dataset Attribute	Details
Total transactions	284,807
Input features	30
Target variable	Class
Fraudulent transactions	492
Legitimate Transactions	284,315

Target classification:

0: Legitimate transaction

1: Fraudulent transaction

Libraries used

- Pandas and NumPy — Data handling
- Scikit-learn — Preprocessing and Logistic Regression
- Matplotlib and Seaborn — Visualization

1. **Dataset Collection**
Kaggle credit card transactions

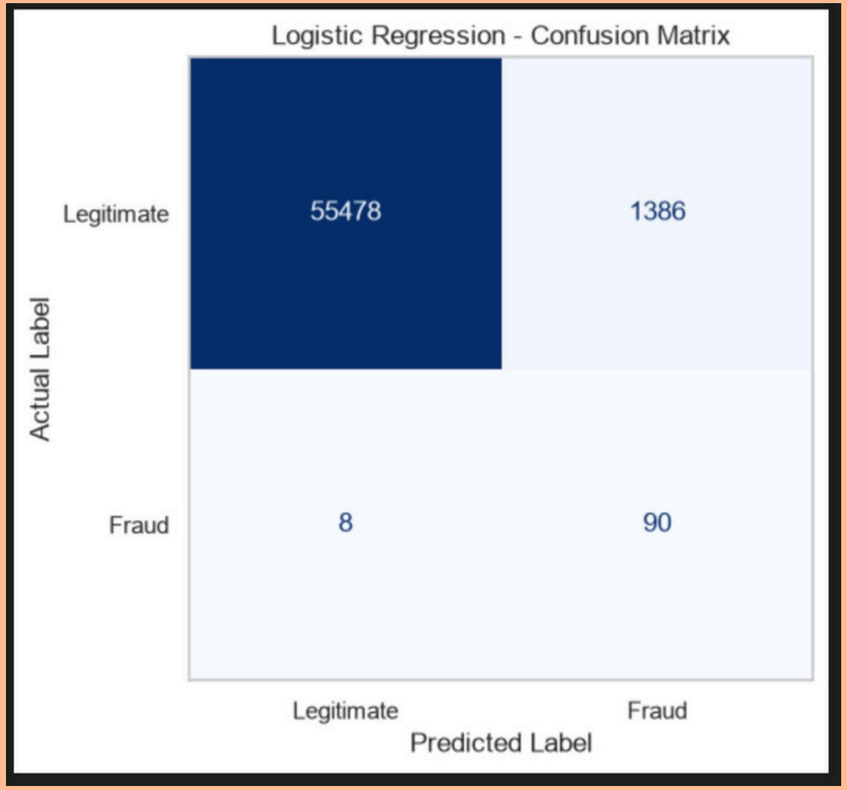
2. **Data Preprocessing**
Duplicate removal • Missing-value handling • Feature selection

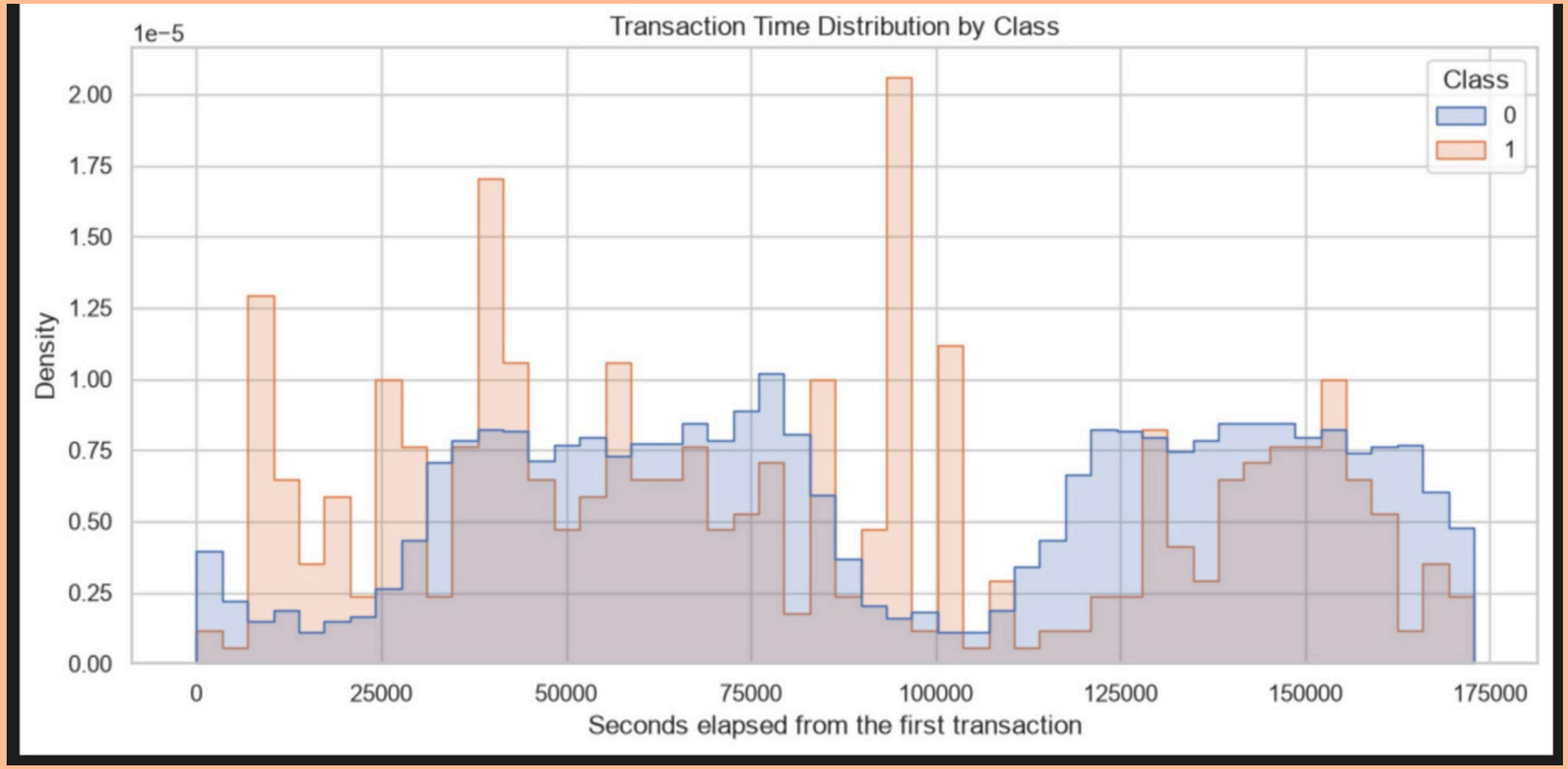
3. **Feature Scaling**
Median imputation • StandardScaler

4. **Model Training**
Logistic Regression

5. **Prediction & Evaluation**
Fraudulent (1) / Legitimate (0) •Performance metrics

Evaluation Results





ACCURACY RATE:
97.55%

PRECISION:
6.10%

RECALL:
91.84%

F1 SCORE:
11.44%

Our Learnings

- Working with a highly imbalanced dataset.
- Data cleaning and feature preprocessing.
- Applying Logistic Regression to a real-world classification problem.
- Evaluating models using precision, recall and F1-score.
- Coordinating with a team in different time zones
- Using Git and GitHub for version control and collaboration

Team Members	Contribution
Ravicharan	Handled documentation of the project, merged and checked the working of the model.
Krishna Kuril	Reviewed code execution, trained the model, and evaluated its performance.
Roshini	Prepared presentation and preprocessed the data.
Sidhartha	Video editing; handled data loading and the visualisation module.

AI-use disclosure

AI tools were used for conceptual clarification, debugging assistance and documentation support. The team reviewed and tested the implementation and took responsibility for the final work.